

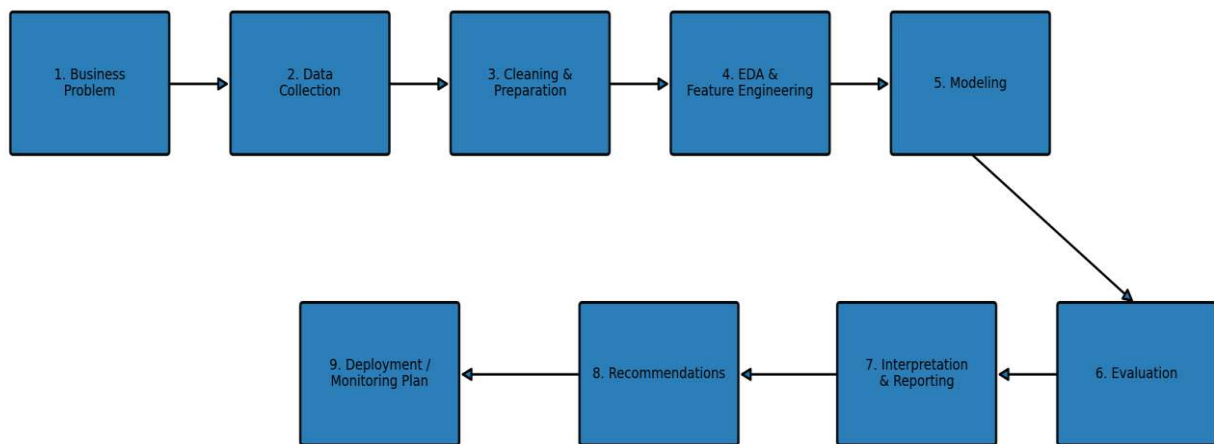
WEEK 1 PROJECT PLAN

Customer Churn Prediction Using Python

Data Science Project Planning and Strategy Design

Hypothetical project

No dataset required for Week 1, Planned effort: 31 hours



Iterative feedback loop: evaluation → feature/model refinement → re-evaluation

Prepared as a strategic planning document for a Python-based data science workflow.

1. Project Background and Introduction

Customer churn refers to customers discontinuing a service or subscription. For a hypothetical telecommunications company, losing customers can reduce recurring revenue and increase the cost of acquiring replacement customers. The proposed project will use Python-based data science methods to identify patterns associated with churn and develop a predictive model that estimates the probability that an existing customer will leave.

The project is intentionally planned without a dataset in Week 1. The focus is on defining the business problem, data requirements, analytical workflow, modeling strategy, evaluation criteria, tools, risks, and expected outcomes so that the later technical implementation can proceed in a controlled and reproducible manner.

Motivation and Problem Justification

- Customer retention is a measurable business problem that can be translated into a binary classification task.
- Historical customer information can potentially reveal relationships between churn and factors such as tenure, contract type, billing behavior, service usage, complaints, and support interactions.
- A predictive approach can help prioritize customers for further review rather than treating every customer as equally likely to churn.
- The project provides a complete learning path covering data preparation, exploratory analysis, feature engineering, machine learning, evaluation, interpretation, and reporting.

2. Project Objectives and Scope

Objectives

1. Define a reproducible Python workflow for customer churn analysis.
2. Identify data fields and potential factors that may be associated with customer churn.
3. Prepare a clean analytical dataset while documenting missing values, duplicates, outliers, and data-quality assumptions.
4. Explore churn patterns through descriptive statistics and visualizations.
5. Design and compare suitable classification models such as Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting.
6. Evaluate model performance using metrics appropriate for a potentially imbalanced binary classification problem.
7. Translate model findings into clear, non-causal business insights and recommendations for further investigation.

In Scope

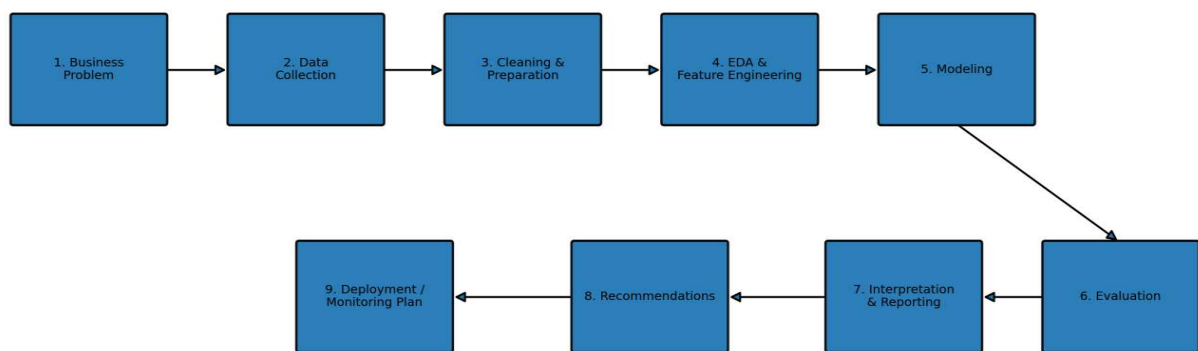
- Customer-level structured/tabular data.
- Data quality assessment and preprocessing.
- Exploratory data analysis and visualization.
- Feature engineering and encoding/scaling where appropriate.
- Baseline and machine-learning classification models.
- Train/validation/test strategy and cross-validation.
- Performance evaluation, error analysis, feature interpretation, and documentation.

Out of Scope

- Collecting real customer data during Week 1.
- Real-time production deployment or automated customer outreach.
- Making causal claims that a feature directly causes churn.
- Using sensitive personal attributes for discriminatory targeting.
- Building a complete commercial retention campaign or financial ROI model without validated business data.

3. Methodology and Strategy

The project will follow an iterative data science lifecycle. Each stage will produce an explicit output that feeds the next stage, while evaluation results may trigger refinement of earlier decisions.



Iterative feedback loop: evaluation → feature/model refinement → re-evaluation

3.1 Problem Definition

Define churn precisely, identify the prediction horizon, specify the unit of analysis, and establish business and technical success criteria.

3.2 Data Collection Strategy

Potential sources include customer master records, subscription/contract details, billing history, service usage, support tickets, complaints, and churn labels. The final schema will be documented before modeling.

3.3 Data Quality and Cleaning

Inspect data types, missingness, duplicate records, inconsistent categories, impossible values, outliers, and target-label quality. Avoid arbitrary deletion; document each rule and its rationale.

3.4 Exploratory Data Analysis

Study target distribution and relationships between churn and candidate variables using summary statistics, histograms, box plots, bar charts, correlation analysis where appropriate, and segment-level comparisons.

3.5 Feature Engineering

Create meaningful variables such as tenure groups, average monthly usage, support-contact frequency, payment behavior indicators, and contract duration. Prevent leakage by using only information available before the prediction point.

3.6 Modeling

Start with Logistic Regression as an interpretable baseline. Compare tree-based models such as Decision Tree, Random Forest, and Gradient Boosting. Use scikit-learn pipelines to keep preprocessing and modeling reproducible.

3.7 Evaluation

Use a held-out test set and cross-validation. Evaluate precision, recall, F1-score, ROC-AUC and, when churn is rare, PR-AUC. Review the confusion matrix and threshold-dependent business trade-offs.

3.8 Interpretation and Reporting

Explain important predictors using model coefficients or feature-importance methods. Clearly distinguish predictive association from causation. Present limitations, assumptions, and recommended next steps.

Proposed Evaluation Logic

The primary metric will be selected after inspecting class balance and the business cost of false negatives versus false positives. Recall is important if missing a likely churner is costly, while precision matters when intervention resources are limited. F1-score can summarize the precision-recall trade-off, and ROC-AUC/PR-AUC can compare ranking quality across thresholds. The final threshold should be chosen using documented business assumptions rather than defaulting automatically to 0.50.

4. Data Requirements and Proposed Schema

A hypothetical dataset could contain the following categories of variables:

Category	Example Variables	Purpose	Potential Issues
Customer profile	Customer ID, region, tenure	Identify records and customer lifecycle stage	Identifiers, missing values
Contract	Plan type, contract length, renewal status	Capture subscription structure	Category inconsistency
Billing	Monthly charge, total charge, payment method	Measure financial behavior	Outliers, skew
Usage	Data/minutes/streaming usage	Measure service engagement	Scale differences
Support	Ticket count, complaint count	Capture service friction	Potential temporal leakage
Target	Churn flag	Binary prediction target	Label quality and class imbalance

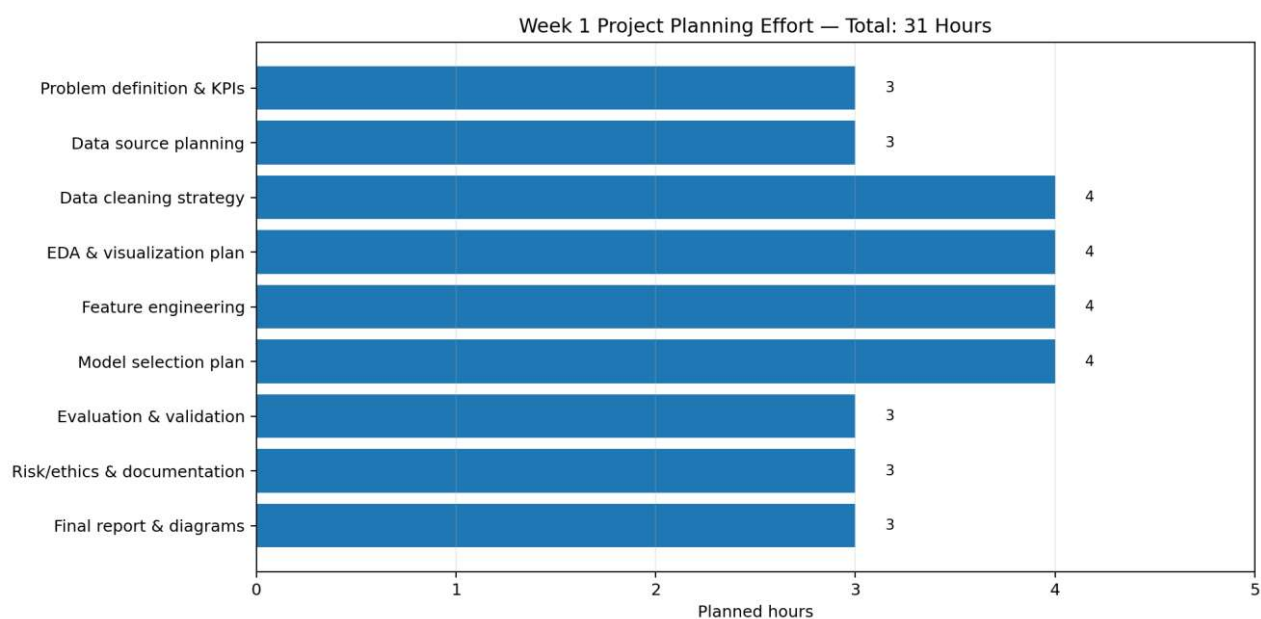
5. Python Tools, Libraries and Working Environment

Tool / Library	Planned Use	Stage
Python 3.x	Core programming language	All stages
Jupyter Notebook / JupyterLab	Exploration, experiments, documentation	EDA / Modeling
pandas	Data loading, cleaning, transformation	Data preparation

NumPy	Numerical operations	Preparation / Modeling
Matplotlib	Core visualizations	EDA / Reporting
Seaborn	Statistical visualizations	EDA
scikit-learn	Preprocessing, models, pipelines, metrics, cross-validation	Modeling / Evaluation
SciPy	Optional statistical tests and numerical utilities	EDA / Analysis
Git / GitHub	Version control and project history	All stages
VS Code	Code organization and development	Implementation

6. Work Plan and 31-Hour Timeline

The Week 1 workload is planned at approximately 31 hours. The allocation is flexible and can be adjusted after the initial project assumptions are reviewed.



Day/Block	Activity	Hours	Deliverable
1	Problem definition & KPIs	3	Problem statement, target definition, success criteria
2	Data source planning	3	Data dictionary, source assumptions, access plan
3	Cleaning strategy	4	Quality checks, missing-value and anomaly rules
4	EDA & visualization plan	4	Questions, plots, segmentation plan
5	Feature engineering	4	Candidate feature list and leakage controls
6	Model selection plan	4	Baseline + candidate models and validation approach
7	Evaluation strategy	3	Metrics, threshold logic, error analysis plan
8	Risk, ethics & documentation	3	Risk register, privacy considerations, assumptions
9	Final report & diagrams	3	Integrated Week 1 document and review

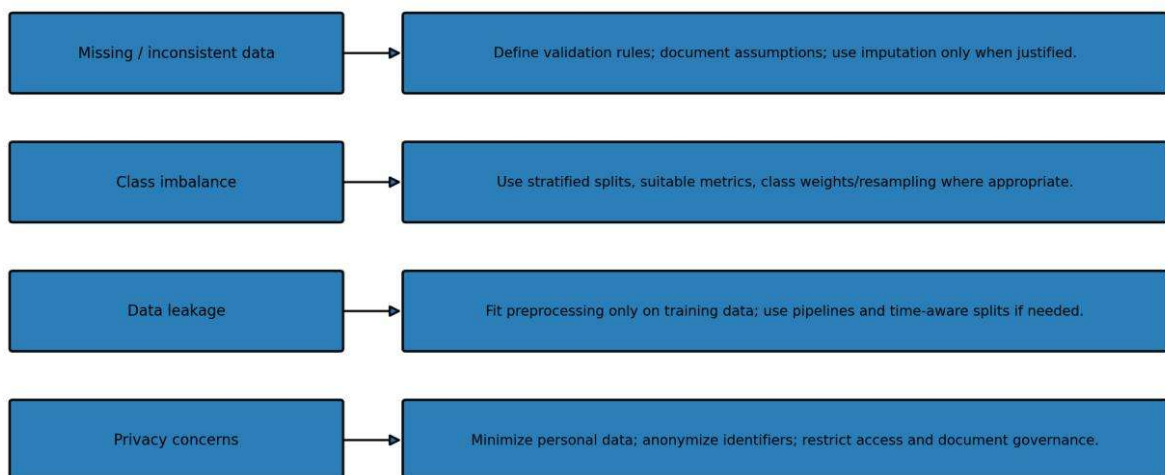
Total		31	
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7. Expected Outcomes and Benefits

- A clearly defined and reproducible project architecture for churn prediction.
- A documented data schema and data-quality checklist ready for implementation.
- An exploratory analysis framework for identifying customer segments associated with different churn rates.
- A baseline model and a structured approach for comparing more advanced classification models.
- A transparent evaluation framework that considers class imbalance, threshold selection, and business trade-offs.
- Actionable analytical outputs such as customer risk segments and key predictive factors, subject to validation on real data.
- A reusable Python project structure that can later be extended to deployment and monitoring.

8. Anticipated Challenges and Mitigation

Anticipated Risks and Mitigation Strategy



9. Quality, Ethics and Reproducibility Strategy

Data privacy

Use the minimum data necessary, remove direct identifiers from modeling tables where possible, control access, and document data-handling assumptions.

Fairness and responsible use

Avoid using sensitive characteristics for harmful targeting. Check whether model errors differ substantially across relevant groups when such analysis is appropriate and lawful.

Reproducibility

Use fixed random seeds where appropriate, version code with Git, record package versions, document preprocessing steps, and keep a clear train/test separation.

Leakage prevention

Ensure that every predictor is available before the defined prediction time. Put transformations inside a pipeline and fit them only on training data.

Interpretability

Prefer understandable baselines and explain important predictors. Do not interpret predictive association as proof of causation.

10. Proposed Project Structure

```
customer_churn_project/
├── data/
│   ├── raw/
│   └── processed/
├── notebooks/
│   ├── 01_data_quality.ipynb
│   ├── 02_eda.ipynb
│   └── 03_modeling.ipynb
├── src/
│   ├── data_cleaning.py
│   ├── features.py
│   ├── modeling.py
│   └── evaluation.py
├── reports/
│   ├── figures/
│   └── final_report.md
├── requirements.txt
└── README.md
```

11. Success Criteria for the Week 1 Plan

- The business problem, prediction target, and scope are unambiguous.
- Required data fields and potential data-quality problems are documented.
- Every major data science stage has a defined strategy and output.
- Model evaluation metrics are justified rather than selected arbitrarily.
- The 30–35 hour workload is explicitly allocated; this plan totals 31 hours.
- Risks, privacy, leakage, and reproducibility are considered before implementation.
- Diagrams and tables make the workflow and schedule understandable without additional explanation.

12. Conclusion

This Week 1 plan establishes a complete strategic foundation for a Python-based customer churn prediction project. The approach begins with a clearly defined business question, identifies the data required, specifies cleaning and exploration procedures, proposes a progression from interpretable baseline models to stronger tree-based models, and defines an evaluation framework that accounts for class imbalance and operational

trade-offs. The 31-hour schedule provides a practical allocation of effort while leaving room for iteration. Once a suitable dataset becomes available, the next phase can translate this plan into notebooks, reusable Python modules, experiments, and a validated model.

Appendix A — One-Page Strategy Summary

Problem	Predict which existing customers may churn.
Type	Supervised binary classification.
Input	Customer, contract, billing, usage, and support features.
Output	Churn probability / predicted churn class.
Baseline	Logistic Regression.
Candidates	Decision Tree, Random Forest, Gradient Boosting.
Key Metrics	Precision, Recall, F1, ROC-AUC, PR-AUC, confusion matrix.
Main Risks	Missing data, imbalance, leakage, privacy, distribution shift.
Planned Effort	31 hours.
Week 1 Deliverable	Complete project plan, strategy diagrams, timeline, tools, outcomes, and risk plan.